

# Dietary intake of nitrate and nitrite and risk of renal cell carcinoma in the NIH-AARP Diet and Health Study

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Background: Nitrate and nitrite are present in many foods and are precursors of N-nitroso compounds, known animal carcinogens and potential human carcinogens. We prospectively investigated the association between nitrate and nitrite intake from dietary sources and risk of renal cell carcinoma (RCC) overall and clear cell and papillary histological subtypes in the NIH-AARP Diet and Health Study. Methods: Nitrate and nitrite intakes were estimated from a 124-item food frequency questionnaire. Over a mean follow-up of 9 years, we identified 1816 RCC cases (n=498, clear cell; n=115, papillary cell) among 491 841 participants. Cox proportional hazard regression was used to estimate hazard ratios (HRs) and 95% confidence intervals (CIs). Results: Individuals in the highest quintile of nitrite intake from animal sources compared with those in the lowest quintile, had an increased risk of total RCC and clear cell subtype (HR=1.28, 95% CI, 1.10–1.49 and HR=1.68, 95% CI, 1.25–2.27, respectively). Nitrite from processed meats and other animal sources were associated with increased clear cell adenocarcinoma risk (HR=1.33, 95% CI, 1.01–1.76 and HR=1.78, 95% CI, 1.34–2.36, respectively). We found no association for nitrite intake from plant sources or nitrate intake overall. Conclusion: Our findings suggest that nitrite from animal sources may increase the risk of RCC, particularly clear cell adenocarcinomas.